Sharp Five-Factor Recht–Ré Lower Bounds in Arbitrary Dimension

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Abstract

We determine a sharp lower Loewner bound for the average of ordered products of five distinct positive semidefinite matrices, for every integer $n \geq 5$ and every finite matrix dimension, under $\sum_i A_i \preceq nI$. Writing $r = n/(n-1)$ and

$$a = \frac{n\big(6n - 9 + \sqrt{20n^2 - 60n + 49}\big)}{16(n-1)(n-2)},$$

the optimal lower constant is $\kappa_5(n) = a^2[r(3-r)a - r^2/2]$. An exact uniform localizer certificate proves the universal inequality; Zhong's known permutation-covariant real n-dimensional family attains it. The proof reduces to four matrix polynomials of sizes $7, 4, 4, 4$ and 104 positive semidefinite Bernstein coefficients over $\mathbb{Q}(\sqrt{5}, \sqrt{249})$, together with two positive scalar blocks. Separate exact implementations check the original free-word identity, the complete all-n representation map, and the positivity data. The strict certificate also yields necessary and sufficient vector conditions for equality. We prove that the minimum attaining dimension is exactly $n-1$ over both fields and give an explicit real tuple in that dimension; every attaining normalized sum saturates the identity on a subspace of dimension at least $n-1$. This answers the five-factor specialization of Lai–Lim's question about the correct lower bound. It does not settle arbitrary product lengths or claim the first ordinary five-factor validity theorem. The manuscript is private and unrefereed; independent AI-agent replay is not human peer review or confirmed priority.

1 Statement and relation to prior work

Let $(n)_5 = n(n-1)(n-2)(n-3)(n-4)$ and define the ordered sum

$$\mathcal{D}_5(A) = \sum_{\substack{(i_1,\dots,i_5)\in\{1,\dots,n\}^5 \\ i_1,\dots,i_5 \text{ distinct}}} A_{i_1}A_{i_2}A_{i_3}A_{i_4}A_{i_5}.$$

Reversal of the five indices shows that this sum is Hermitian.

Theorem 1. *For every integer $n \geq 5$, let $A_1, \dots, A_n$ be positive semidefinite real symmetric or complex Hermitian matrices of any common finite dimension. If $\sum_i A_i \preceq nI$, then*

$$\frac{\mathcal{D}_5(A)}{(n)_5} \succeq -\kappa_5(n)I, \qquad \kappa_5(n) = a^2\left[r(3-r)a - \frac{r^2}{2}\right], \tag{1}$$

where

$$r = \frac{n}{n-1}, \qquad \Delta = \sqrt{20n^2 - 60n + 49}, \qquad a = \frac{n(6n - 9 + \Delta)}{16(n-1)(n-2)}. \tag{2}$$

For every such n, a real n-dimensional tuple with $\sum_i A_i = nI$ attains equality on a nonzero vector. Thus $\kappa_5(n)$ is the exact dimension-unrestricted lower constant over either field.

The universal part of the proof includes finite exact algebraic checks specified in Sections 3–6. Section 2 proves attainment directly. In raw-sum normalization, set $C_5(n) = (n)_5\kappa_5(n)$; then

$$C_5(5) = \frac{3125(9033 + 581\sqrt{249})}{393216}, \qquad \lim_{n\to\infty} \kappa_5(n) = \frac{11 + 5\sqrt{5}}{64}.$$

The result is a lower endpoint, not a comparison with $(n^{-1}\sum_i A_i)^5$, and not by itself an upper norm bound. Section 7 gives vector conditions for equality and its minimum dimension, rather than a classification of all tuples up to unitary equivalence.

Recht and Ré proposed the noncommutative AM–GM problem [1]. Lai and Lim [2] disproved the original conjecture and developed its noncommutative sum-of-squares formulation. Their Section 6 asks, "What is the correct lower bound?" Theorem 1 answers that question for product length five. Their numerical raw-sum value at $(n, m) = (5, 5)$ is prior evidence for this endpoint, not a new disproof here. Zhong's Section 1.1 and Appendix A [4] already supply the permutation-covariant family and recurrence used below, building on De Sa's examples [3]. The new certificate matches the optimized family value universally; the family and recurrence are not claimed as new.

The public candidate repository of `ipitchford` [5] already claims the ordinary five-factor upper bound for $n \geq 5$ and lower bound for $n \geq 6$, with failure at $n = 5$. Its word-orbit and symmetry-block certificate architecture is reused and credited. Our target is the exact lower constant rather than the unit bound. The inspected source is pinned to commit `df3cb842014210eba99d1971be48aa079367feac`. The literature audit is bounded: the full theorem of Liu's CISS 2025 paper [6] was inaccessible through the checked lawful routes. Its accessible abstract does not substitute for its exact theorem or exclude overlap. Priority remains unconfirmed, independently of the mathematical certificate checks.

2 The known family attains the constant

This section is a self-contained specialization of Zhong's family and recurrence [4, Section 1.1 and Appendix A]. In $\mathbb{R}^n$, put $e = n^{-1/2}(1, \ldots, 1)^T$, $P = I - ee^T$, and

$$z_i = \sqrt{r}\left(e_i - \frac{1}{n}(1, \ldots, 1)^T\right).$$

Then $z_i \perp e$, $\|z_i\| = 1$, $z_i^T z_j = -1/(n-1)$ for $i \neq j$, $\sum_i z_i = 0$, and $\sum_i z_i z_i^T = rP$. Let $d = n - 1 - (n-2)a$ and define

$$A_i = a(P - z_i z_i^T) + (e + \sqrt{d}\, z_i)(e + \sqrt{d}\, z_i)^T. \tag{3}$$

These are PSD once $a, d > 0$. To check this for every $n \geq 5$, write $s = 2 - r \geq 3/4$, $b = 3 - r$, and observe that a is the larger root of

$$16sa^2 - 6rba + r^2 = 0. \tag{4}$$

For $1 < r \leq 5/4$, this quadratic has two positive roots; its vertex $3rb/(16s)$ is at most $5/8$. Its value at 1 is $7r^2 - 34r + 32 \geq 7/16 > 0$. Therefore $0 < a < 1$, and $d > n - 1 - (n-2) = 1$. Summing (3) gives

$$\sum_i A_i = nee^T + [na + r(d-a)]P = nI.$$

For any ordered distinct-label word let $t_k = e^T A_{i_k} \cdots A_{i_1} e$, with $t_0 = t_1 = 1$. Write the intermediate vector as $t_k e + \sum_{j=1}^{k} g_{j,k} z_{i_j}$ and $h_k = \sum_j g_{j,k}$. For a fresh label i,

$$A_i e = e + \sqrt{d}\, z_i, \qquad A_i z_j = a z_j - \frac{\sqrt{d}}{n-1} e - \frac{d-a}{n-1} z_i \quad (j \neq i).$$

Consequently

$$\begin{pmatrix} t_{k+1} \\ h_{k+1} \end{pmatrix} = \begin{pmatrix} 1 & -\sqrt{d}/(n-1) \\ \sqrt{d} & a - (d-a)/(n-1) \end{pmatrix} \begin{pmatrix} t_k \\ h_k \end{pmatrix}.$$

The transfer matrix has trace $2a$ and determinant ar. Thus

$$t_{k+2} = 2a t_{k+1} - ar t_k, \qquad t_5 = 8(2-r)a^4 - 4r(3-r)a^3 + r^2 a^2. \tag{5}$$

Every distinct word has the same e-to-e coefficient. Permutation covariance makes $\mathcal{D}_5(A)e$ a multiple of e, so $\mathcal{D}_5(A)e = (n)_5 t_5 e$. Using (4),

$$t_5 = a^2 \left[\frac{r^2}{2} - r(3-r)a \right] = -\kappa_5(n).$$

The stationary quadratic evaluated at $r/[2(3-r)]$ equals $r^2[4(2-r)/(3-r)^2 - 2] < 0$; hence the larger root exceeds that point and $\kappa_5(n) > 0$. This establishes the claimed attained value in both the real and complex settings. No assumption that this family is universally extremal is used: that conclusion follows from the certificate.

3 A free-polynomial localizer certificate

Use labels $0, \ldots, n-1$ in the certificate. Let $\mathcal{W}_n$ be the list consisting of the empty word, the linear words, and all ordered quadratic words, in that order and lexicographic order within each length. Its real span has orthonormal word basis and dimension $1 + n + n^2$. For a word w, $w(A)^*$ reverses its letters; it does not commute them.

Let G be a symmetric Gram matrix indexed by $\mathcal{W}_n$ and invariant under permutations fixing label 0. Define G_i by relabelling 0 as i. Let H be a symmetric Gram matrix invariant under all label permutations. The certificate identity is

$$\begin{aligned} C_5(n) I + \mathcal{D}_5(A) = &\sum_i \sum_{w,v \in \mathcal{W}_n} (G_i)_{w,v}\, w(A)^* A_i v(A) \\ &+ \sum_{w,v \in \mathcal{W}_n} H_{w,v}\, w(A)^* \left(nI - \sum_i A_i \right) v(A). \end{aligned} \tag{6}$$

If $G, H \succeq 0$, factoring these scalar matrices expresses the right side as sums of $p(A)^* A_i p(A)$ and $q(A)^*(nI - \sum_i A_i) q(A)$. This proves positivity for the full domain $\sum_i A_i \preceq nI$ in any finite dimension. The equality tuple is used to choose certificate coordinates; the universal proof never replaces the slack by zero.

3.1 Canonical coordinates and the word identity

For two words w, v, encode their concatenation with a separator between them, canonically renaming new letters in order of first appearance. For G, reserve 0 for the marked label and name the others $1, 2, \ldots$; for H, start with 0. Take the lexicographically smaller code obtained from (w, v) and (v, w),

with separator -1. Do not reverse either word within this code. This describes exactly simultaneous label invariance and Gram symmetry. There are 59 marked and 22 unmarked orbits. Enumerating pairs in $\mathcal{W}_6 \times \mathcal{W}_6$ and retaining each code on its first occurrence fixes coordinates $z_0, \ldots, z_{80}$, with the marked orbits first. At most four active labels occur in a pair, so this indexing is valid for all $n \geq 5$.

The exact rational functions $z_c(u)$ are the ordered list `orbit_functions` in the frozen `CERTIFICATE.json`; Appendix A specifies its 18 free coordinates and the rational parameter u is defined below. All 81 functions are machine-readable proof data, not numerical approximations.

To check (6), take a concrete target word and sum over its splits into a prefix, one inserted letter, and a suffix, with each outer length at most two. The relevant G_i entry has the reversed prefix as its left word; subtract the corresponding H entry. For each split without an inserted letter, add n times that H entry. The required coefficient is $C_5(n)$ for the empty word, 1 for a degree-five word with five distinct labels, and 0 otherwise. All words through degree five are covered by their 76 equality patterns. These coefficient equations have been checked as identities in $\mathbb{Q}(u)$, not by evaluating finitely many n. The target uses the raw ordered sum: there is no factor $5!$ or $(n)_5$ inside (6). If $X_i = A_i/n$, the raw lower constant becomes $C_5(n)/n^5$ on $\sum_i X_i \preceq I$.

4 The complete invariant representation

We give the representation argument because checking a few contractions of a Gram matrix is insufficient without completeness. Let q be the number of permuted letters: $q = n$ for H, and $q = n - 1$ for G. In the latter case 0 is fixed and all sums in this section use the other letters. In both cases $q \geq 4$.

The invariant vectors for H, in order, are

$$1, \quad \sum_i x_i, \quad \sum_i x_i^2, \quad \sum_{i \neq j} x_i x_j.$$

For G they are

$$1, \ x_0, \ \sum_i x_i, \ x_0^2, \ \sum_i x_0 x_i, \ \sum_i x_i x_0, \ \sum_i x_i^2, \ \sum_{i \neq j} x_i x_j.$$

These independent vectors have disjoint word supports and span the entire fixed subspace. Contracting the Gram matrices against them defines H_0 of size 4 and G_0 of size 8.

For $c \in \mathbb{R}^q$ with $\sum_i c_i = 0$, define

$$L(c) = \sum_i c_i x_i, \qquad Q(c) = \sum_i c_i x_i^2,$$
$$P(c) = \sum_{i \neq j} (c_i + c_j) x_i x_j, \qquad M(c) = \sum_{i \neq j} (c_i - c_j) x_i x_j.$$

These give four copies of the standard representation for H, in the order L, Q, P, M. For G use the six copies

$$L(c), \quad \sum_i c_i x_0 x_i, \quad \sum_i c_i x_i x_0, \quad Q(c), P(c), M(c).$$

They are orthogonal by support or by symmetry versus skew symmetry. Their squared norm multipliers relative to $\|c\|^2$ are 1 except for P, M, where they are $2(q-2), 2q$. Contract all copies against the same $c = e_0 - e_1$ for H, or $e_1 - e_2$ for G, obtaining H_1 of size 4 and G_1 of size 6.

The remaining pair spaces are the symmetric zero-diagonal arrays with zero row sums, denoted U_q, and the skew arrays with zero row sums, denoted V_q. Their dimensions are

$$\dim U_q = \frac{q(q-3)}{2}, \qquad \dim V_q = \frac{(q-1)(q-2)}{2}.$$

Indeed, the symmetric row-sum map has rank q for $q \geq 3$ and the skew row-sum map has rank $q-1$; their complementary subspaces are exactly the pair copies already displayed.

For completeness, the permutation module on unordered pairs has a three-dimensional commutant: an equivariant entry depends only on intersection size $0, 1, 2$. It contains one trivial and one standard summand, so its remaining nonzero summand U_q has scalar commutant and is absolutely irreducible. On skew pairs, the image of $e_1 \wedge e_2$ determines an equivariant map. Invariance under the remaining permutations and antisymmetry under (1 2) leave two directions:

$$e_1 \wedge e_2, \qquad \sum_{j>2} (e_1 \wedge e_j - e_2 \wedge e_j).$$

Thus that commutant has dimension two. Removing its standard copy leaves the absolutely irreducible V_q. The types are inequivalent by dimension except at $q = 4$, where the standard and V_4 both have dimension three. Their transposition characters are 1 and -1, respectively, so they remain inequivalent. This covers the marked $\mathfrak{S}_4$ boundary at $n = 5$.

Use the symmetric switch

$$x_i x_j + x_j x_i + x_k x_l + x_l x_k - x_i x_k - x_k x_i - x_j x_l - x_l x_j$$

and the skew triangle

$$x_i x_j + x_j x_k + x_k x_i - x_j x_i - x_k x_j - x_i x_k$$

as representatives of U_q, V_q. Their squared norms are $8, 6$. For H take $(i, j, k, l) = (0, 1, 2, 3)$; for G take $(1, 2, 3, 4)$. The resulting scalar contractions are H_2, H_3, G_2, G_3.

Lemma 2. *For every integer $n \geq 5$, $G \succeq 0$ if and only if $G_0, G_1, G_2, G_3 \succeq 0$, and similarly for H.*

Proof. The dimensions exhaust the word space:

$$\begin{aligned} H: &\quad 4 + 4(q-1) + q(q-3)/2 + (q-1)(q-2)/2 = 1 + n + n^2, \quad q = n, \\ G: &\quad 8 + 6(q-1) + q(q-3)/2 + (q-1)(q-2)/2 = 1 + n + n^2, \quad q = n - 1. \end{aligned}$$

On each standard copy, equivariant bilinear forms are scalar multiples of the standard inner product. Evaluation at the common c multiplies the multiplicity form by $\|c\|^2 = 2$, with invertible positive diagonal changes for the embedding norms. On U_q, V_q, the contractions are positive multiples $8, 6$ of their scalar eigenvalues. Absolute irreducibility and inequivalence leave no untested coupling. The same real Gram matrices apply to complex Hermitian inputs. □

This also gives the symmetric invariant coordinate counts $8 \cdot 9/2 + 6 \cdot 7/2 + 2 = 59$ and $4 \cdot 5/2 + 4 \cdot 5/2 + 2 = 22$.

4.1 Exact construction of the block map

For representatives $R_{B,i}$ just specified, each contracted entry is

$$B_{ij}(n) = \sum_{c=0}^{80} L_{B,ij,c}(n) z_c(u), \qquad L_{B,ij,c}(n) = \sum_{\substack{w,v \in \mathcal{W}_n \\ \mathrm{orbit}(w,v)=c}} R_{B,i}(w) R_{B,j}(v), \tag{7}$$

with the appropriate marked or unmarked coordinate offset. The left word in this contraction is not reversed; reversal occurs only in the free-polynomial coefficient extraction. Fix the representative's anchor labels, including the mark for G. There are respectively $1, 3, 5, 4$ anchors for G_0, G_1, G_2, G_3, and $0, 2, 4, 3$ for H_0, H_1, H_2, H_3. A word pair contains at most four fresh labels. A canonical fresh-label pattern with k distinct labels has $(n - \text{number of anchors})_k$ injective realizations. This derives (7) as a polynomial of degree at most four in n. Impossible patterns automatically have falling factorial zero.

An independent implementation reconstructed the integer count maps at $n = 5, 6, 7, 8, 9$ and used exact degree-four interpolation. The preceding degree bound makes this a proof of the count maps, not empirical extrapolation of rational candidate entries. After substituting $n(u)$ and the 81 functions z_c, all 81 upper-triangular block entries were checked as rational identities: $36+21+10+10+4 = 81$. Thus the positivity data belong to the actual Gram matrices in (6), rather than to an unrelated saved block array.

5 The rational chart and kernel reductions

Put

$$E = 16u^2 - 12u + 1, \qquad N = 32u^2 - 18u + 1,$$
$$n = \frac{N}{E}, \quad a = \frac{N}{16u - 6}, \quad u_0 = \frac{3 + \sqrt{5}}{8}, \quad u_5 = \frac{21 + \sqrt{249}}{48}. \tag{8}$$

On $u_0 < u \le u_5$, $E > 0$ and

$$n'(u) = -\frac{2(48u^2 - 16u + 3)}{E^2} < 0, \qquad n(u_5) = 5, \qquad \lim_{u \downarrow u_0} n(u) = \infty.$$

The numerator quadratic is positive by its negative discriminant. The inverse branch is

$$u = \frac{6n - 9 + \Delta}{16(n - 2)}, \qquad a = \frac{nu}{n - 1}.$$

It satisfies (4), equivalently $16(n-1)(n-2)a^2 - 6n(2n-3)a + n^2 = 0$. Every finite integer $n \ge 5$ therefore occurs exactly once. Real-parameter polynomial positivity below is only a proof device; the matrix-tuple theorem has integer n.

The denominator ledger is exhausted by the linear factors with roots

$$0,\ 1,\ 1/16,\ 1/2,\ 1/6,\ 1/8,\ 3/8$$

and the three quadratics

$$u^2 - \frac{15u}{16} + \frac{3}{32}, \quad u^2 - \frac{17u}{16} + \frac{9}{64}, \quad u^2 - \frac{3u}{4} + \frac{1}{16}. \tag{9}$$

The exact verifier checks divisibility into these factors with constant remainder. They have no physical zero: $u_0 > 1/2$ and $u_5 < 37/48$; the first two quadratic pairs are $(15 \pm \sqrt{129})/32$ and $(17 \pm \sqrt{145})/32$, with small roots below $1/2$ and large roots above $13/16$ and $29/32$. The final large root is exactly the excluded u_0. These observations justify every finite specialization, including $n = 5$.

Let $\ell = n-(n-2)a$, $\beta = (n-2)a/(n-1)$, and $\gamma = a[2(n-2)a-n]/(n-1)$. The exact blocks satisfy

$$G_0 q_T = 0, \qquad q_T = (1, \ell, \beta, \ell^2, \beta\ell, \beta\ell, \beta\ell, \gamma)^T, \tag{10}$$

$$G_1 q_S = 0, \qquad q_S = (1, a, a, \ell, a, 0)^T,$$

$$G_1 k = 0, \qquad k = (0, n-1, -(n-1), 0, 0, 1)^T. \tag{11}$$

The first two equations comprise the original fourteen kernel rows; the last contributes six additional checked rows. Here

$$2(n-2)a - n = \frac{n(\Delta - 2n - 1)}{8(n-1)} > 0, \quad \Delta^2 - (2n+1)^2 = 16(n-1)(n-3) > 0,$$

so $\gamma > 0$. Delete coordinate 7 from G_0 (zero-based indexing). For G_1, the rows $4, 5$ of the two kernel columns q_S, k form $\mathrm{diag}(a, 1)$; delete those coordinates.

Lemma 3. *If a Hermitian matrix M annihilates the columns of K and the rows K_J at a deleted coordinate set J form an invertible square matrix, then $M \succeq 0$ if and only if the retained principal block is PSD.*

Proof. For any vector x, let $y = x - KK_J^{-1}x_J$. It has zero coordinates on J and $x^*Mx = y^*My$. The converse is principal compression. □

Thus the remaining matrix sizes are $7, 4, 4, 4$; no exact-rank assumption at a special parameter is required. The scalar blocks, not imported from a lower-degree certificate, are

$$G_2 = \frac{-1024u^2 + 1088u - 144}{8u - 3}, \qquad H_2 = \frac{-1024u^2 + 960u - 96}{8u - 3}, \tag{12}$$

$$G_3 = 0, \qquad H_3 = 0.$$

The first two are positive: their numerators are -1024 times the second and first quadratics in (9), respectively, which are negative on the interval. Also $8u - 3 > 0$ and $G_2 - H_2 = 16$.

6 The finite positivity certificate

Here B denotes one of the retained blocks $G_0[0{:}6], G_1[0{:}3], H_0, H_1$. For the exponent vector h_B in Table 1, set

$$D_B(n) = \mathrm{diag}(n^{h_{B,i}-2}), \qquad P_B(u) = d_B(u)^2 D_B(n(u)) B(u) D_B(n(u)). \tag{13}$$

The recorded general construction permits division by a common power E^{v_B} after squaring the denominator; the accepted data have $v_B = 0$ for all four blocks. Thus (13) is the exact conversion used here. All entries of D_B are nonzero and d_B has no physical zero, so it is an invertible congruence times a positive scalar. It preserves PSD in both directions.

With $v = u - 1/16$, $w = u - 1/2$, $z = u - 3/8$, the monic denominators are

$$d_{G_0} = uv^3w^3z^4, \quad d_{G_1} = vwz^3, \quad d_{H_0} = uv^4w^4z^4, \quad d_{H_1} = uv^3w^3z^2.$$

The polynomial conversion is verified entry by entry as an exact rational identity, including the indicated degrees.

B	size	h_B	$\deg P_B$	Bernstein degree q_B
G_0	7	$(0,1,0,2,1,1,1)$	24	28
G_1	4	$(1,2,2,2)$	12	16
H_0	4	$(0,0,1,0)$	28	32
H_1	4	$(1,2,1,1)$	20	24

Table 1: Exact normalization and the four coefficient families.

Set $t = (u - u_0)/(u_5 - u_0)$. Every P_B has the exact expansion

$$P_B(u_0 + (u_5 - u_0)t) = \sum_{j=0}^{q_B} \binom{q_B}{j} t^j (1-t)^{q_B - j} C_{B,j}, \qquad C_{B,j} \in \mathbb{Q}(\sqrt{5}, \sqrt{249})^{s_B \times s_B}. \tag{14}$$

There are $29 + 17 + 33 + 25 = 104$ coefficient matrices. Each is symmetric PSD and the four final coefficients C_{B,q_B} are PD. The exact matrices and their rational-field LDL^T factors are stored in `CERTIFICATE.json`. Its SHA-256 is

```
f4727ed67e787acb10208ca8fe7ea7a6c
787cebfb117faf77df8647022843225.
```

This hash identifies the proof data; a hash alone is not a positivity argument. The checks establishing their asserted properties are specified next and have completed independent replays.

6.1 Exact arithmetic and deterministic acceptance

Represent each field element by four rational coordinates

$$A + B\sqrt{5} + C\sqrt{249} + D\sqrt{1245} = (A + B\sqrt{5}) + (C + D\sqrt{5})\sqrt{249},$$

with both radicals positive. Addition and multiplication follow their quadratic relations. For $a + b\sqrt{5}$ with opposite nonzero signs, the sign is that of a if $a^2 - 5b^2 > 0$, and that of b if it is negative. Zero or matching signs are immediate. For $x + y\sqrt{249}$ with $x, y \in \mathbb{Q}(\sqrt{5})$ of opposite signs, compare $x^2 - 249y^2$ in the smaller field in the same way. These norm tests give exact real signs, with no tolerance or floating eigenvalue. The rational basis is independent because 5, 249, and 1245 are nonsquares, so coordinate equality is exact.

For each supplied factorization, the independent checker reconstructs $C = LDL^T$, verifies symmetry and nonnegative diagonal entries of D, and requires strictly positive diagonal entries for the final coefficient. During factor construction, a zero pivot is allowed only with an entirely zero remaining residual row. The replay can check the supplied factors without repeating elimination. A factorization with a negative pivot or failed identity is rejected.

To verify (14), first compose the rational polynomial with $u_0 + (u_5 - u_0)t$ exactly. If its power coefficients are p_i, the degree-q Bernstein coefficients must satisfy

$$C_k = \sum_{i=0}^{k} \frac{\binom{k}{i}}{\binom{q}{i}} p_i, \qquad p_i = \binom{q}{i} \sum_{k=0}^{i} (-1)^{i-k} \binom{i}{k} C_k.$$

Both are matrix identities. The independent replay used Horner composition and forward differences, separate from the discovery implementation, and checked every power coefficient including those beyond the true degree. The verifier also reconstructs all 81 orbit functions, all 76 word equations, all 20 kernel rows, the four scalar blocks, and all normalization and denominator identities. The separate count-map replay described after (7) closes the remaining representation-map dependency.

Proposition 4. *The frozen rational functions define PSD Gram matrices G, H satisfying (6) for every integer $n \geq 5$.*

Proof. For $0 < t \leq 1$, every scalar Bernstein weight in (14) is nonnegative, while the final weight is $t^{q_B} > 0$. Since $C_{B,q_B} \succ 0$, all four $P_B(u)$ are PD on the physical interval. Equation (13) gives PD of the retained blocks. Lemma 3, the verified kernels, and (12) give PSD of all eight full blocks. Their maps are the contractions of the orbit Gram matrices, so Lemma 2 gives $G, H \succeq 0$. The exact 76 coefficient identities give (6). No positivity assumption is needed at the excluded infinite-n endpoint to infer positivity at finite n. □

Factoring the Gram matrices now proves the universal inequality in Theorem 1; Section 2 proves its sharpness. Numerical optimization was used to discover a rational candidate, but solver status, a floating margin, and rounding accuracy have no role in this implication. Verification evaluates the fixed rational data only.

7 Equality and the minimum attaining dimension

The main inequality and its original n-dimensional sharpness witness do not depend on this section. We now use the *strict* positivity proved in Proposition 4, not merely the known kernel inclusions. Retain the raw parameter $\ell = n - (n-2)a$ and put

$$\begin{gathered} X_i = A_i/n, \qquad S = \sum_i X_i \preceq I, \\ \lambda = \ell/n, \qquad h = a/n = \frac{1-\lambda}{n-2}, \qquad b = \frac{1}{n-1}. \end{gathered} \tag{15}$$

Theorem 5. *Under the hypotheses of Theorem 1, a unit vector e satisfies $\mathcal{D}_5(A)e = -C_5(n)e$ if and only if*

$$Se = e, \qquad SX_ie = X_ie, \tag{16}$$

$$X_i^2 e = \lambda X_i e, \tag{17}$$

$$(X_iX_j + X_jX_i)e = 2h(X_ie + X_je - be) \qquad (i \neq j). \tag{18}$$

For every attaining tuple, $\dim\ker(I - S) \geq n - 1$. *The minimum ambient attaining dimension is exactly $n - 1$, over both $\mathbb{R}$ and $\mathbb{C}$. In that minimum dimension, every attaining tuple satisfies $S = I$ on its whole space.*

The proof uses the five-factor Gram kernels directly. It does not import a four-factor polynomial identity, nor assert a quantitative stability result or confirmed priority for the equality theorem. There is a direct earlier analogue at length two: Zhao's Theorem 5.1 [7] classifies real two-factor equality through a common reducing canonical block and proves minimum attaining dimension n. The present result concerns a different endpoint and product length, and gives vector conditions rather than that stronger form of tuple classification. Here "minimum dimension" always means dimension for attaining the exact sharp endpoint. Lai–Lim's separate question about the smallest dimension supporting *any* violation of the ordinary five-factor inequality [2, Section 3] is not settled here.

7.1 Full kernel coordinates from strict block positivity

For finite n, the retained G_0, G_1 blocks are PD, so their full kernels have dimensions exactly one and two, spanned by q_T and q_S, k in (10)–(11). The blocks H_0, H_1, G_2, H_2 are strictly positive and $G_3 = H_3 = 0$. Hence the only other Gram kernels are the skew zero-row-sum pair modules. These assertions concern finite n; no rank conclusion at u_0 is needed.

The contracted kernels lift to the original word coordinates without an inverse norm factor. Indeed, if V embeds the representatives of a block, $M = V^*GV$, $G \succeq 0$, and $Mq = 0$, then $(Vq)^*G(Vq) = 0$, whence $GVq = 0$. Use all zero-sum standard weight vectors in this statement. Completeness of the representation and the strict ranks show that the lifted vectors exhaust the kernel.

Fix a mark k. A vector-valued array $y = (y_w)_{w\in\mathcal{W}_n}$ belongs to $\ker G_k$ tensored with the ambient space precisely when it has the following form. There are vectors f, ξ_j, ζ_j for $j \neq k$, with $\sum_{j\neq k}\xi_j = \sum_{j\neq k}\zeta_j = 0$, and an active skew array Z_{ij} with zero row sums, such that

$$\begin{aligned}
(y_\varnothing, y_k, y_j) &= (f, \ell f, \beta f + \xi_j),\\
(y_{kk}, y_{jj}) &= (\ell^2 f, \beta\ell f + \ell\xi_j),\\
y_{kj} &= \beta\ell f + a\xi_j + (n-1)\zeta_j,\\
y_{jk} &= \beta\ell f + a\xi_j - (n-1)\zeta_j,\\
y_{ij} &= \gamma f + a(\xi_i + \xi_j) + (\zeta_i - \zeta_j) + Z_{ij}.
\end{aligned} \tag{19}$$

Every displayed single index j is active, so $j \neq k$; the last line uses $i \neq j$ with $i, j \neq k$; also $Z_{ii} = 0$. The coefficients follow by inserting q_T, q_S, k into the original representatives in Section 4. In particular the standard skew representative contributes $\zeta_i - \zeta_j$, while the marked copies contribute $\pm(n-1)\zeta_j$. Similarly, $\ker H$ consists exactly of arrays supported on distinct quadratic words, skew in their two labels and with zero row sums.

7.2 Necessity on the full subunital domain

Write $R = \sum_i A_i$ and $T = nI - R \succeq 0$. Equality in (6) forces the vector arrays

$$\begin{aligned}
y_w &= A_k^{1/2} w(A) e \in \ker G_k \otimes \mathcal{H},\\
z_w &= T^{1/2} w(A) e \in \ker H \otimes \mathcal{H},
\end{aligned} \tag{20}$$

where $\mathcal{H}$ is the real or complex ambient space. The constant and linear coordinates of z vanish, so $Te = TA_ie = 0$. This is exactly (16). The marked linear coordinate in (19) gives $A_k^{1/2}A_ke = \ell A_k^{1/2}e$; multiplication by $A_k^{1/2}$ yields $A_k^2 e = \ell A_k e$, proving (17).

For $i \neq j$, define the raw symmetric defect

$$F_{ij} = A_iA_j + A_jA_i - 2a(A_i + A_j - rI).$$

The identities

$$\beta(\ell - a) = a(\ell - r), \qquad \gamma = a(2\beta - r)$$

follow from the parameter definitions. If $k = i$ or $k = j$, the sum of the two marked/active coordinates in (19), using the first identity, gives $A_k^{1/2}F_{ij}e = 0$. If k is different from both, the active-pair coordinates give the same conclusion by the second identity. The constant, linear and symmetric quadratic coordinates of z vanish, so $T^{1/2}F_{ij}e = 0$ as well. Therefore

$$n\|F_{ij}e\|^2 = \sum_k \|A_k^{1/2}F_{ij}e\|^2 + \|T^{1/2}F_{ij}e\|^2 = 0.$$

Dividing $F_{ij}e = 0$ by n^2 gives (18). The slack term is essential here: $RF_{ij}e = 0$ alone would not justify deleting the defect on a subunital tuple.

7.3 Sufficiency, including the skew coordinates

Assume (16)–(18), expressed also in raw variables. Fix k and set

$$f = A_k^{1/2}e, \qquad \xi_j = A_k^{1/2}A_j e - \beta f,$$
$$\zeta_j = \frac{A_k^{1/2}[A_k, A_j]e}{2(n-1)} \quad (j \neq k).$$

The equality $A_k(A_k - \ell I)e = 0$ and $\ker A_k = \ker A_k^{1/2}$ imply $y_k = \ell f$ and $y_{kk} = \ell^2 f$. Saturation gives $Re = ne$ and $[A_i, R]e = 0$, hence $\sum \xi_j = \sum \zeta_j = 0$. Also $y_{jj} = \ell y_j$. The symmetric parts of all pair coordinates in (19) follow from $F_{ij}e = 0$ and the two parameter identities above.

For active labels define $d_{ij} = (y_{ij} - y_{ji})/2$. Their row sums are

$$\sum_{j \neq k} d_{ij} = \frac{1}{2}A_k^{1/2}[A_i, R - A_k]e = \frac{1}{2}A_k^{1/2}[A_k, A_i]e = (n-1)\zeta_i.$$

Thus $Z_{ij} = d_{ij} - (\zeta_i - \zeta_j)$ is skew with zero row sums. This completes every coordinate of (19), so each G-localizer term has zero quadratic form at e.

For the slack array, the constant and linear coordinates vanish by saturation, the diagonal quadratic ones by the quadratic condition, and the symmetric pair coordinates by $F_{ij}e = 0$. Finally

$$\sum_j z_{ij} = T^{1/2}A_i Re = nT^{1/2}A_i e = 0.$$

It lies in the exact skew kernel of H. All terms of the positive identity have zero quadratic form. Since its left side is PSD, it annihilates e, proving the converse in Theorem 5.

7.4 The dimension obstruction over both fields

Let $v_i = X_i e$ and $p_i = \langle e, X_i e\rangle \geq 0$. The equality conditions give

$$\sum_i v_i = e, \qquad \sum_i p_i = 1, \qquad \|v_i\|^2 = \lambda p_i.$$

Testing (18) against e first gives $\mathrm{Re}\langle v_i, v_j\rangle = h(p_i + p_j - b)$. In the complex case, testing against v_i and using $X_i v_i = \lambda v_i$ gives

$$(\lambda - 2h)\langle v_i, v_j\rangle = 2h(\lambda - b)p_i - \langle v_i, X_j v_i\rangle \in \mathbb{R}.$$

By Section 2, $\delta := \lambda - 2h = 1 - a > 0$. Hence the Gram matrix is real and equals

$$M = (\langle v_i, v_j\rangle) = \mathrm{diag}(\delta p_i + hb) + h(p\mathbf{1}^T + \mathbf{1}p^T - b\mathbf{1}\mathbf{1}^T).$$

For any nonzero real or complex z with $\sum_i z_i = 0$,

$$z^* M z = \sum_i (\delta p_i + hb)|z_i|^2 > 0.$$

Thus $\mathrm{rank}\, M \geq n - 1$. Each v_i belongs to $\ker(I - S)$ by saturation, proving $\dim \ker(I - S) \geq n - 1$ over either field.

7.5 A real attaining tuple in dimension $n-1$

Put $m = n-1$ and work in $\mathbb{R}^m$. Choose a unit vector e and unit regular-simplex vectors $z_1, \dots, z_m$ in $e^\perp$. Thus $\sum_j z_j = 0$, $z_i^T z_j = -1/(m-1)$ for $i \neq j$, and $\sum_j z_j z_j^T = mP/(m-1)$, where $P = I - ee^T$. Set

$$\alpha = 1/m = b, \qquad d = \lambda - \alpha, \qquad \rho = (1-\lambda)/m.$$

Because $0 < a < 1$, one has $\lambda > 2/n > 1/(n-1)$, so $d > 0$; also $\rho, h > 0$. Define

$$\begin{aligned} X_j &= \rho(P - z_j z_j^T) + (\sqrt{\alpha}e + \sqrt{d}\, z_j)(\sqrt{\alpha}e + \sqrt{d}\, z_j)^T \quad (1 \le j \le m), \\ X_n &= 2hP. \end{aligned} \tag{21}$$

All matrices are PSD. Their mixed terms cancel; the first m matrices have ee^T coefficient one and P coefficient

$$\frac{m(m-2)\rho + md}{m-1} = 1 - 2h.$$

Thus $\sum_i X_i = I$. For $j \le m$, $v_j = \alpha e + \sqrt{\alpha d}\, z_j$ lies in the rank-one block with eigenvalue $\alpha + d = \lambda$. Hence $X_j^2 e = \lambda v_j$, also valid for $X_n e = 0$. For distinct active labels,

$$X_i z_j = \rho z_j - \frac{\sqrt{\alpha d}}{m-1} e - \frac{d-\rho}{m-1} z_i.$$

The identities $\alpha - d/(m-1) = h$ and $m\rho/(m-1) = h$ give

$$(X_i X_j + X_j X_i)e = 2\alpha h e + 2h\sqrt{\alpha d}(z_i + z_j) = 2h(v_i + v_j - be).$$

For the final label, $(X_n X_j + X_j X_n)e = 2h\sqrt{\alpha d}\, z_j = 2h(v_j - be)$. All equality conditions hold. Their proved sufficiency and the scaling $A_i = nX_i$ give attainment of the same sharp five-factor endpoint in dimension $n-1$. This completes Theorem 5.

8 Reproducibility, scope, and review status

This LaTeX source compiles without external inputs. The expanded 81 rational orbit functions, affine table, normalization polynomials, and 104 factorized coefficient matrices form a finite machine-readable supplement. The companion private directory is `overnight-ncagm-m5-all-n-private-package-20261019`; its `README.md` specifies the data manifest. With CPython 3.12 and the pinned SymPy 1.14.0 dependency already installed, its offline interface is

```
python3.12 -I verify.py
```

No optimizer, network access, API credential, or original workspace is required. The two stages use the byte-identical frozen certificate at `stages/psd_words/INPUT-CERTIFICATE.json` and `stages/block_map/CERTIFICATE.json`. The wrapper's only adaptation of a mathematical worker removes three original absolute-environment guard lines in its temporary copy; the exact diff is included. A separate portable validation completed both stages with all exact checks passing, no solver calls, and no mathematical core changes. Its receipt is `portable.stdout.json` in the companion `overnight-ncagm-m5-portable-validation-20261019` directory. The source manifest beside this manuscript identifies that receipt and the frozen research files replayed.

Two complementary independent exact implementations have completed: the first checks the 81 functions, 76 original word patterns, 20 kernel rows, scalar conditions, polynomial transforms

and all 104 coefficient factorizations; the second reconstructs the complete all-n block map and checks its 81 upper-triangular identities. They share exact rational-polynomial infrastructure but do not import the discovery worker. The count-map replay uses exact integer counts and the proved degree bound, not sampled positivity. Analytical reviews separately establish representation completeness, physical-interval signs, the operator bridge, and attainment. The main acceptance record is `ACCEPTANCE.json`, hash

`c64dd723abacd28724f3a8472dee2df3f7`
`caa725b91ceca04cf3d826d26084ff.`

The equality extension was subsequently accepted after two separate analytical reviews of its full kernels, slack argument, complex Gram obstruction and smaller construction. Its separate source and acceptance hashes are included in `SOURCE-CLAIM-MANIFEST.json`; the original main-theorem acceptance is unchanged. The source preceding that extension is retained as `pre-equality.tex`.

The proof and software were developed and audited by Codex AI agents in a private campaign. "Independent" here means a separately implemented check by another agent, not human refereeing or formal verification in a proof assistant. No human author identity, credential, endorsement, adoption, or employment consequence is asserted. The manuscript has not been published or sent to the cited authors. It settles only the stated five-factor lower-envelope problem; it does not establish a universal extremizer for other product lengths, the optimal general quadratic restoration threshold, or an algorithmic convergence comparison. Broader priority checking and human mathematical review remain open tasks.

A Exact free-coordinate specification

The 18 free orbit columns F_j, their degrees d_j, and integers p_j, b_j in Table 2 define

$$z_{F_j}(n) = n^{4-d_j}\left[\frac{p_j}{2^{32}5^{4-d_j}} + \left(\frac{1}{n} - \frac{1}{5}\right)\frac{b_j}{2^{32}}\right]. \tag{22}$$

Thus $z_{F_j}(5) = p_j/2^{32}$ is the fixed exact five-label seed. These are the final accepted slopes, not the earlier rejected candidate. The remaining coordinates are the exact rational strings in the hash-bound `CERTIFICATE.json`; equivalently they are obtained from `INPUT-AFFINE.json`'s 81×19 `solution_matrix_rows` applied to $(z_{F_0}, \ldots, z_{F_{17}}, 1)^T$. Its hash is

`7f8c881a0f58aa0d7e20f5e0b002b5c5`
`d5b493299e8cb772d9eadd79f526a639.`

The canonical indexing in Section 3 fixes the meaning of every column, so no unspecified ordering convention is required. The validity check reconstructs the target identity from these functions; it does not rely on the history of how the affine equations were solved.

In JSON, each field coordinate is a list of four rational strings. Their basis order is

$$(1, \sqrt{5}, \sqrt{249}, \sqrt{1245}).$$

The certificate's `coefficient_certificates` entries contain `name`, `bernstein_degree`, and an ordered coefficient list. Each list item has its index `k`, exact `matrix`, and `LDL`, including the lower factor and diagonal pivots. The `POLYNOMIAL-CHECKPOINT.json` entries specify h_B, d_B, v_B and the polynomial affine rows before insertion of the fixed slopes. All polynomial strings use the single variable `u` over $\mathbb{Q}$. This specification, the listed finite data, and the exact arithmetic rules in Section 6 determine a deterministic certificate check without an optimizer or a choice of test points.

j	F_j	d_j	$p_j = 2^{32} z_{F_j}(5)$	b_j
0	17	3	-20817478263	88271623635
1	25	3	-18396979216	84921120947
2	61	2	11706295450	-9365392863
3	62	2	12981031495	-6837242709
4	64	2	75433334948	-81291345767
5	65	3	-24441626383	77610325069
6	66	3	-17077590016	71145486647
7	68	3	-12903045670	70652260805
8	69	3	-13157619467	57507367161
9	70	4	5969730599	-76348992586
10	71	4	2907939245	-55936656701
11	72	4	4181582164	-48912958583
12	73	4	2235060079	-67273566954
13	74	4	1929660160	-48211383911
14	76	4	3769223925	-104625015253
15	78	4	3242886733	-81769581127
16	79	4	5296781463	-98155140573
17	80	4	2541005637	-57879970852

Table 2: Fixed dyadic seed and slope data in canonical orbit order.

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